

National University of Computer and Emerging Sciences, Lahore Campus
Quiz1 [BS(CY): Section 3A] Fall 2026

Cyber Security (CY2004)

Quiz Date: September , 2026

Total Marks: 20

Duration: 20 -Minutes

Name ----- Roll #----- Section -----

Q #1: Encircle the correct option. Cutting/overwriting is not allowed.

[11 Marks]

- I. An attacker intercepts a bank transfer and silently changes the destination account number, without ever viewing the sender's personal details. Which CIA goal is primarily violated?
A) Integrity B) Confidentiality C) Availability D) Authentication
- II. During a Black Friday sale, an e-commerce site crashes because too many shoppers try to check out simultaneously. Which CIA goal has failed?
A) Integrity B) Confidentiality **C) Availability** D) Authentication
- III. A company completely air-gaps (disconnects from the internet) a laptop containing sensitive R&D data. What is the most likely trade-off?
A) All three goals improve simultaneously B) Availability increases; integrity suffers
C) Confidentiality increases; availability may suffer D) Integrity increases; confidentiality suffers
- IV. A bank's fraud detection system flags a login because the typing rhythm doesn't match the user's usual pattern, even though the correct password was entered. What kind of authentication factor is being used here?
A) Token: something you have B) Knowledge: something you know
C) Behavioral biometric: something you are D) Physiological biometric: something you are
- V. At border control, an officer first checks your passport details against the travel database, then scans your fingerprint to confirm you are the person named on that passport. What is the fingerprint scan an example of?
A) Identification **B) Verification** C) Access control D) Non-repudiation
- VI. Select the correct true/false combination for the following two statements: (i) A system can lose integrity while still fully preserving confidentiality, (ii) A hotel key card is an example of the 'something you know' authentication factor.
A) True, True **B) True, False** C) False, False D) False, False
- VII. A university stores students' examination results in a database. An unauthorized employee manages to view the results but does not change or delete anything. Which security property has primarily been violated?
A) Integrity **B) Confidentiality** C) Availability D) Authentication
- VIII. An employee receives an email containing a malicious attachment. The employee opens the attachment, allowing malware to execute on the computer. In this scenario, what is the most appropriate description of the attack vector?
A) The computer of the employee B) The malware itself
C) The malicious email attachment D) The confidential company database
- IX. A system administrator gives every employee administrator-level privileges because "they may need them someday." An ordinary employee's account is later compromised by an attacker. Which fundamental security principle was violated?
A) Defense through obscurity B) Non-repudiation
C) Availability **D) Principle of least privilege**
- X. Select the correct true/false combination for the following two statements: (i) A password and a PIN normally represent two different authentication factors because both are examples of *something you know*., (ii) A vulnerability is a potential weakness that may be exploited, whereas an attack is an intended unauthorized action against a system or asset.
A) True, True B) True, False C) False, False D) False, False
- XI. A student successfully logs into a university portal but is denied access to the administrator's database; which security function makes this decision?
A) Identification **B) Authorization** C) Accounting D) Authentication

Q #2: Answer the following questions:

[3+3+3 = 9 Marks]

A) A hospital's patient database is fully encrypted and highly available, but a nurse silently edits a patient's medication dosage without authorization. Explain which CIA goal is compromised and why the other two remain intact.

Answer: Integrity is compromised, since the data (dosage) was modified without authorization. Confidentiality remains intact because no unauthorized disclosure/viewing occurred — encryption still protects the data from being read by outsiders. Availability remains intact because the system is still accessible to authorized users; the record can still be reached and read (albeit now incorrect).

B) Explain the difference between identification and verification using a workplace login example (not from the lecture) of your own.

Answer: **Identification** is the one-to-many step of establishing who someone claims to be, e.g., typing in an employee ID or username, which the system looks up among all registered employees. **Verification** is the one-to-one step of confirming that claim is genuine — e.g., the system then checks the password or a second factor tied specifically to that employee ID to confirm the person really is who they claim to be.

C) A company's server is running an outdated application containing a known security weakness. An attacker uses specially crafted input to take advantage of the weakness and gain unauthorized access.

Explain what vulnerability and exploit are in this scenario, and identify the attack vector used by the attacker.

Answer: The **vulnerability** is the known security weakness in the outdated application. The **exploit** is the specially crafted input or technique used by the attacker to take advantage of that weakness. The **attack vector** is the method or path through which the attacker sends the malicious input to reach the server.